Neuron, Volume *74*

Supplemental Information

The Local Transcriptome in the Synaptic Neuropil Revealed by Deep Sequencing and High-Resolution Imaging

Iván J. Cajigas, Georgi Tushev, Tristan J. Will, Susanne tom Dieck, Nicole Fuerst, and Erin M. Schuman

Supplemental Information Inventory

Supplemental Figures

Figure S1, related to Figure 3. 454 sequencing statistics and comparison of our data set to recently published studies.

Figure S2, related to Figure 4. Dissociated glia cultures possess few neuronal cells.

Figure S3, related to Figure 5. Contribution of each reference list to the filtering process.

Figure S4, related to Figure 6. Raw images from high-resolution *in situ* hybridization experiments.

Figure S5, related to Figure 6. Processing of high-resolution *in situ* hybridization images and relative occupancy of *Dlg4* mRNA particles in the soma and dendrite.

Figure S6, related to Figure 6. High-resolution fluorescent *in situ* hybridization detection of mRNA transcripts in hippocampal dendrites.

Figure S7, related to Figures 5 and 6. High-resolution in situ hybridization control experiments and verification of filtering process.

Supplemental Tables (available separately online)

Table S1: List of all transcripts identified in Neuropil layer of CA1 Hippocampus after 454 sequencing

Table S2: Gene ontology analysis of the Neuropil identified transcripts (8379)

Table S3: List of all transcripts identified in Somata layer of CA1

Hippocampus after 454 sequencing

Table S4: List of transcripts from Zhong et al., Poon et al. and Lein et al. that were not detected in our analyzed data set.

Table S5: Quantitative Nanostring data for transcripts (293) measured in Neuropil and Somata layers

Table S6: Transcripts validated by real time PCR

Table S7: Glia-enriched and Somata-enriched transcripts from Figures 4B and 4D

Table S8: List of all subtracted transcripts (5829) from Neuropil list including the group membership of each filtered list

Table S9: List of all subtracted transcripts (4803) from Somata list including the group membership of each filter list

Table S10: Filtered Neuropil list (2550)

Table S11: Gene Ontology analysis of Neuropil filtered transcripts (2550)

Table S12: Filtered Somata list (3251)

Table S13: CA1 transcriptome (3508) identified by 454 sequencing, filtered out glia, interneurons and endothelial transcripts

Table S14: Process localized mRNAs (87) validated by FISH in previous studies

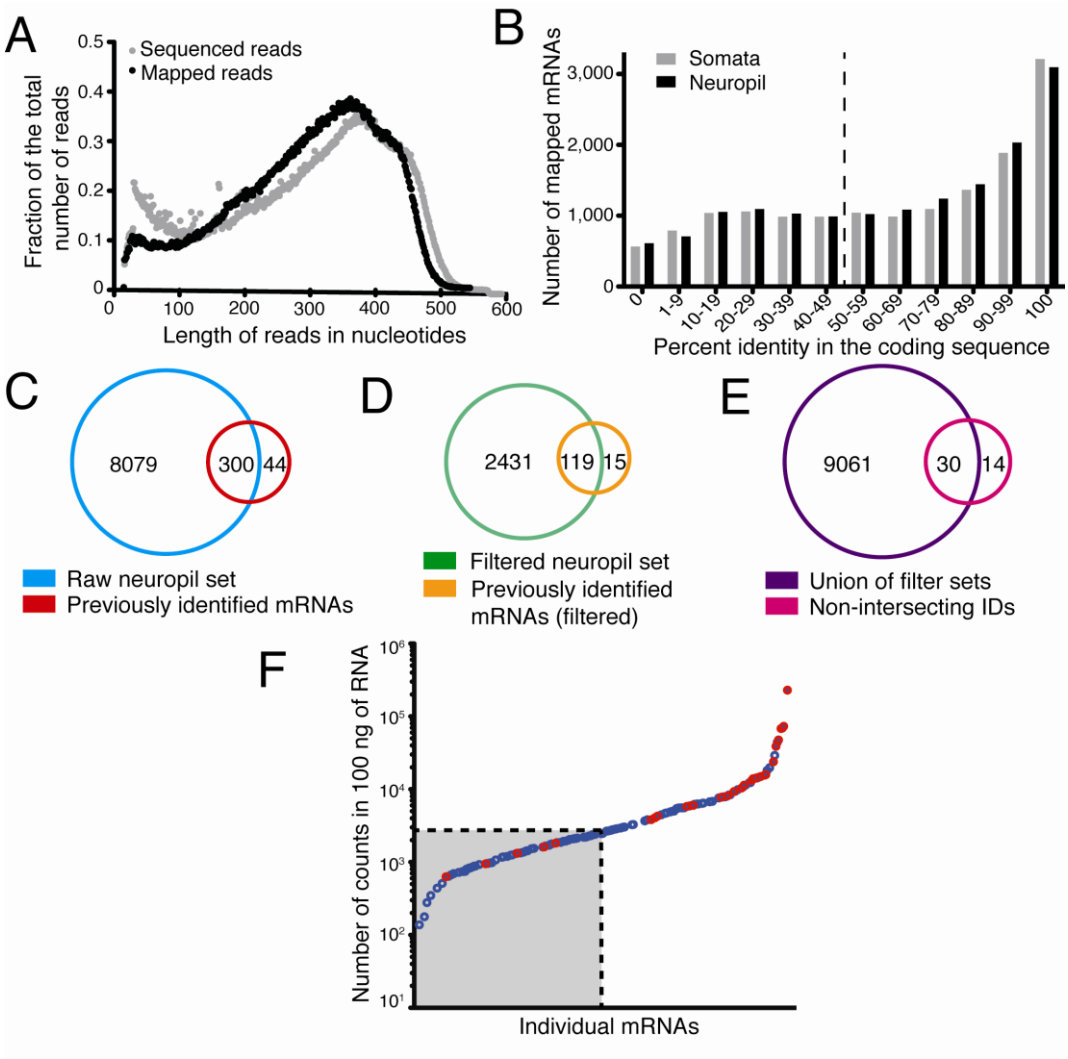


Figure S1, related to Figure 3. 454 sequencing statistics and comparison of our data set to recently published studies

(A) Scatter plots showing the frequency of reads as a function of length (nucleotides) for all sequenced reads (gray circles) and reads that mapped to the transcriptome (black circles).

(B) Bar graph representing the percent identity of mapped reads to the coding sequences of the rat transcriptome. mRNAs in which the entire coding sequence was covered greater than or equal to 50% (right side of the dashed line) were annotated and analysed further.

(C) Venn diagram showing the overlap between our raw neuropil data set and the combined mRNAs identified in three recently published neuropil data sets (Zhong et al., 2006; Poon et al., 2006; Lein et al., 2007). The overlap between the two groups is signficant: Fisher's exact test p= 2.48E-51***. Note that if we do not include the criteria (see methods) for 50% coverage in the codings sequence for a transcript to be included in our data set, the overlap increases to 97.4%.

(D) Venn diagram showing the overlap between our filtered neuropil data set and the filtered list of mRNAs identified in previous studies. The overlap between the two groups is signficant: Fisher's exact test p= 1.31E-81***

(E) Venn diagram of non-intersecting IDs (the 44 transcripts that were not found in our raw neuropilset- see A) from previous studies compared to the union of all filter sets. The overlap between the two groups is signficant: Fisher's exact test p= 0.0134*

(F) Scatter plot showing the abundance of some mRNAs from our filtered data set and some from the filtered list of mRNAs identified in previous studies (red circles). Around 83% of the mRNAs previously detected are in the most abundant half of our filtered set. The remaining 17% are depicted in the gray box and represent the least abundant half of our filtered set.

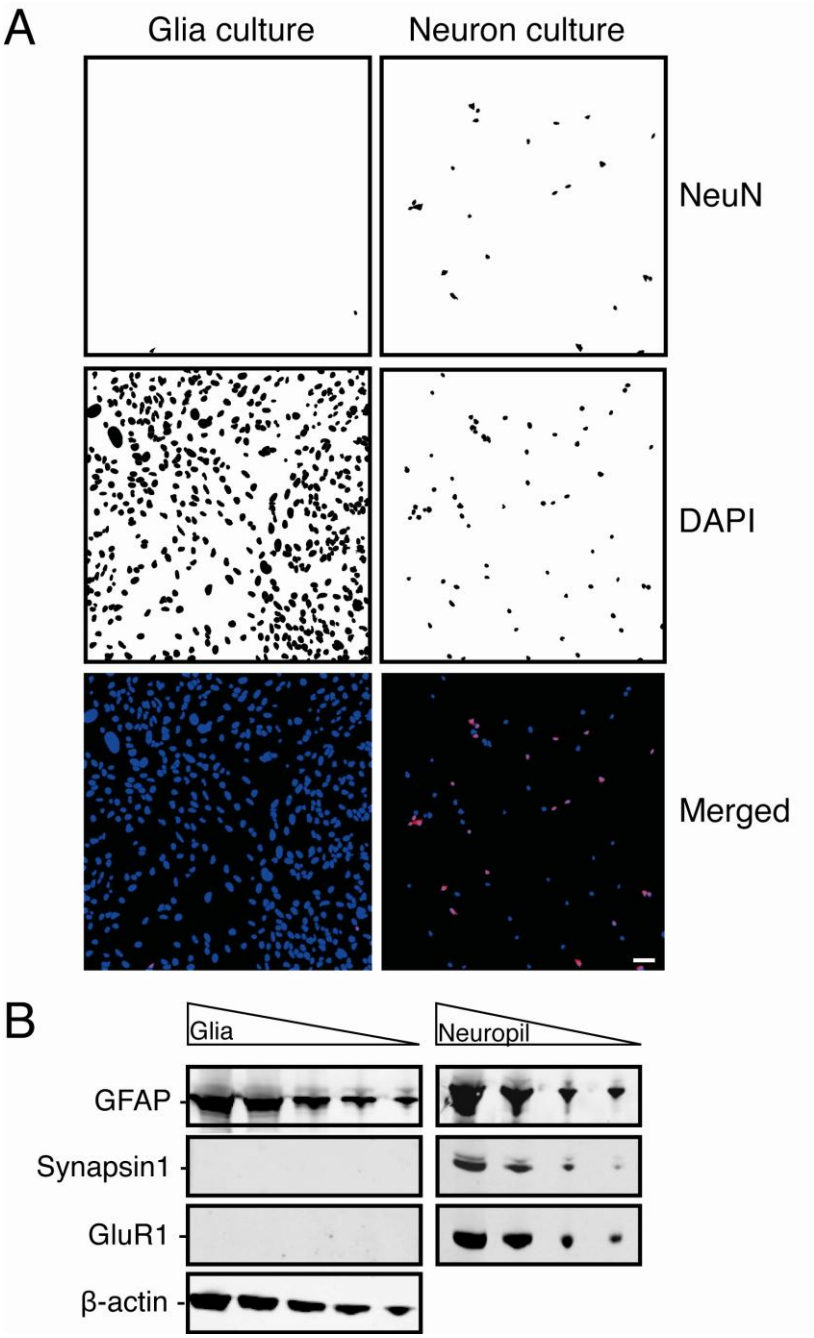


Figure S2, related to Figure 4. Dissociated glia cultures possess few neuronal cells.
(A) A 1 week-old glia culture (left) or standard hippocampal neuronal culture (right) (21 DIV) was immunostained for NeuN (black or red in top and bottom images, respectively) in order to detect neurons. The complete population of cells (glia + neurons) was revealed by staining the nucleus with DAPI (blue). Scale bar = 50 microns.
(B) Lysates (20 - 1 mg) from both glia cell cultures and the microdissected CA1 neuropil were prepared and analyzed for the presence of glial (GFAP) and synaptic (Synapsin and GluR1) markers by Western blot.

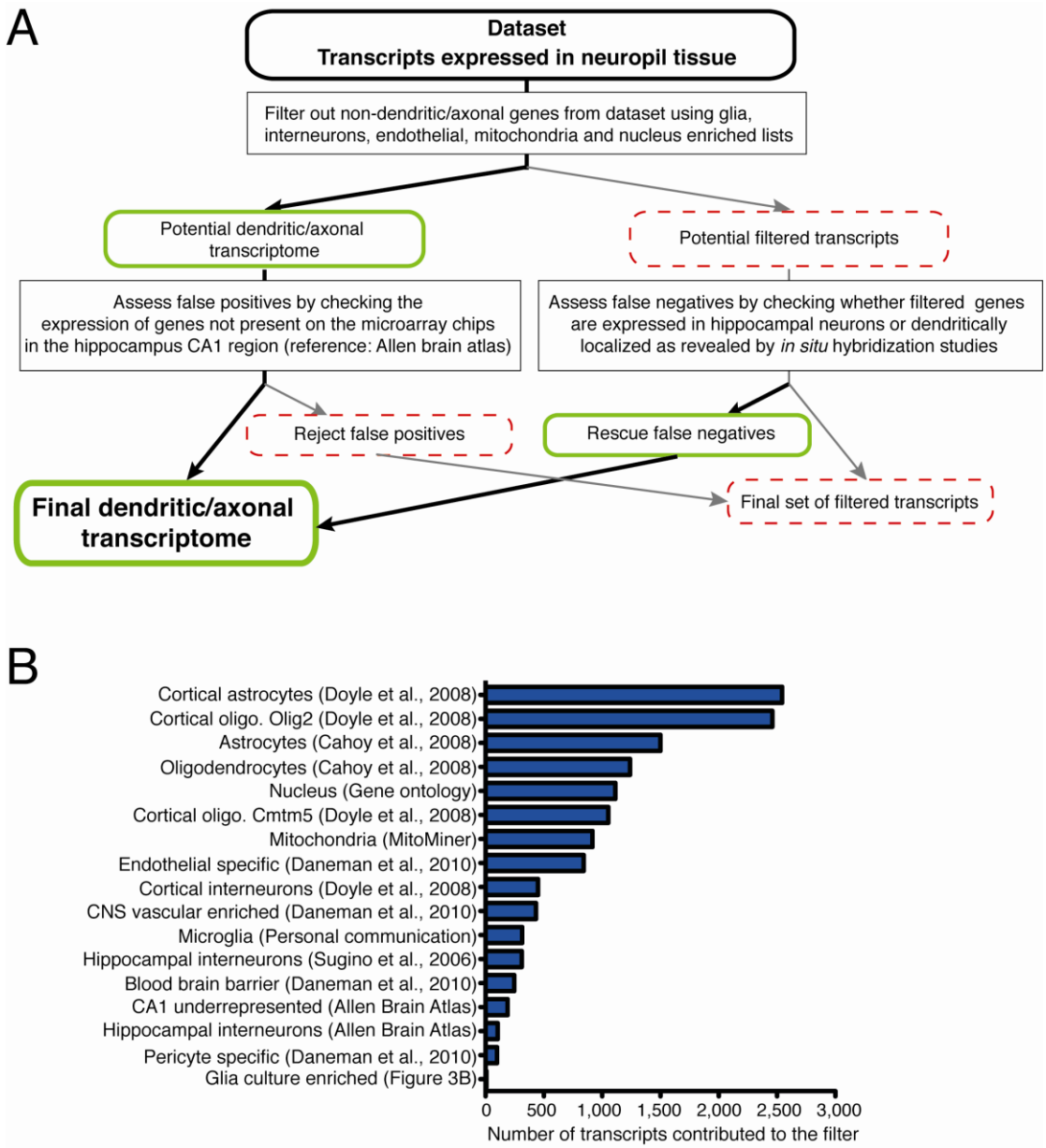


Figure S3, related to Figure 5. Contribution of each reference list to the filtering process.

(A) A data set was generated from sequencing of RNA samples obtained from the neuropil of area CA1 of the rat hippocampus. The data set was "cleaned" with a filter containing gene lists from glia, interneurons, endothelial, mitochondria and nucleus. Two groups were derived from the initial cleaning (Potential dendritic/axonal transcriptome and Potential filtered transcripts). In order to identify false positives, we checked for genes that were not represented in the Affymetrix chips used in the published studies (see B) that we used to filter the data. These genes were removed from the neuropil transcriptome. On the other hand, we confirmed false negatives by checking the overlap between the "Potential filtered transcripts list" with a hippocampal neuron transcriptome (Sugino et al., 2005) and a list containing genes previously validated by *in situ* hybridization (Table S14). The genes found in the sesources were rescued from the false negative list and added back to the "Final dendritic/axonal transcriptome".

(B) Bar graph showing the genes from different reference lists that overlapped with the neuropil set. Note that some genes are present in more than one list.

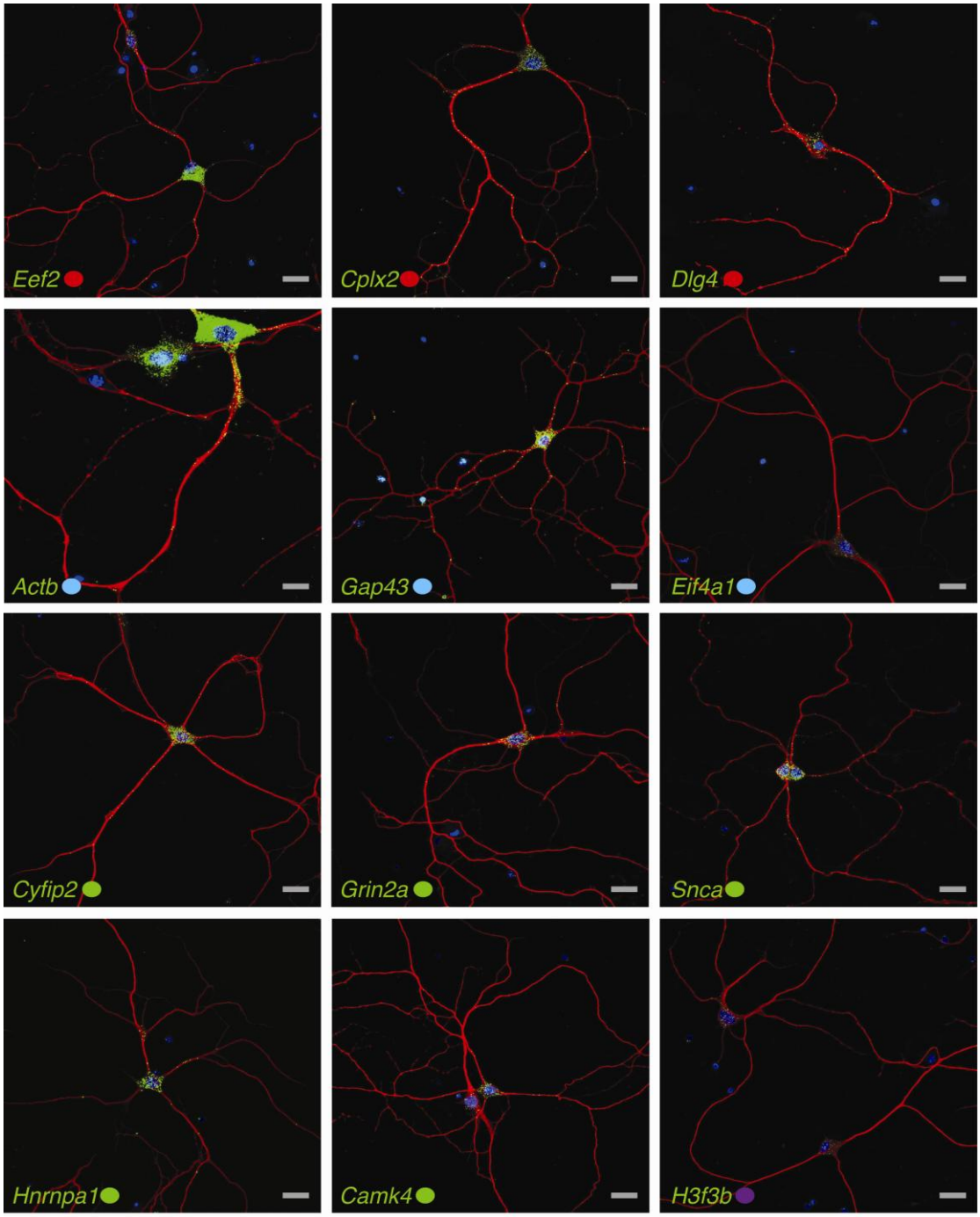
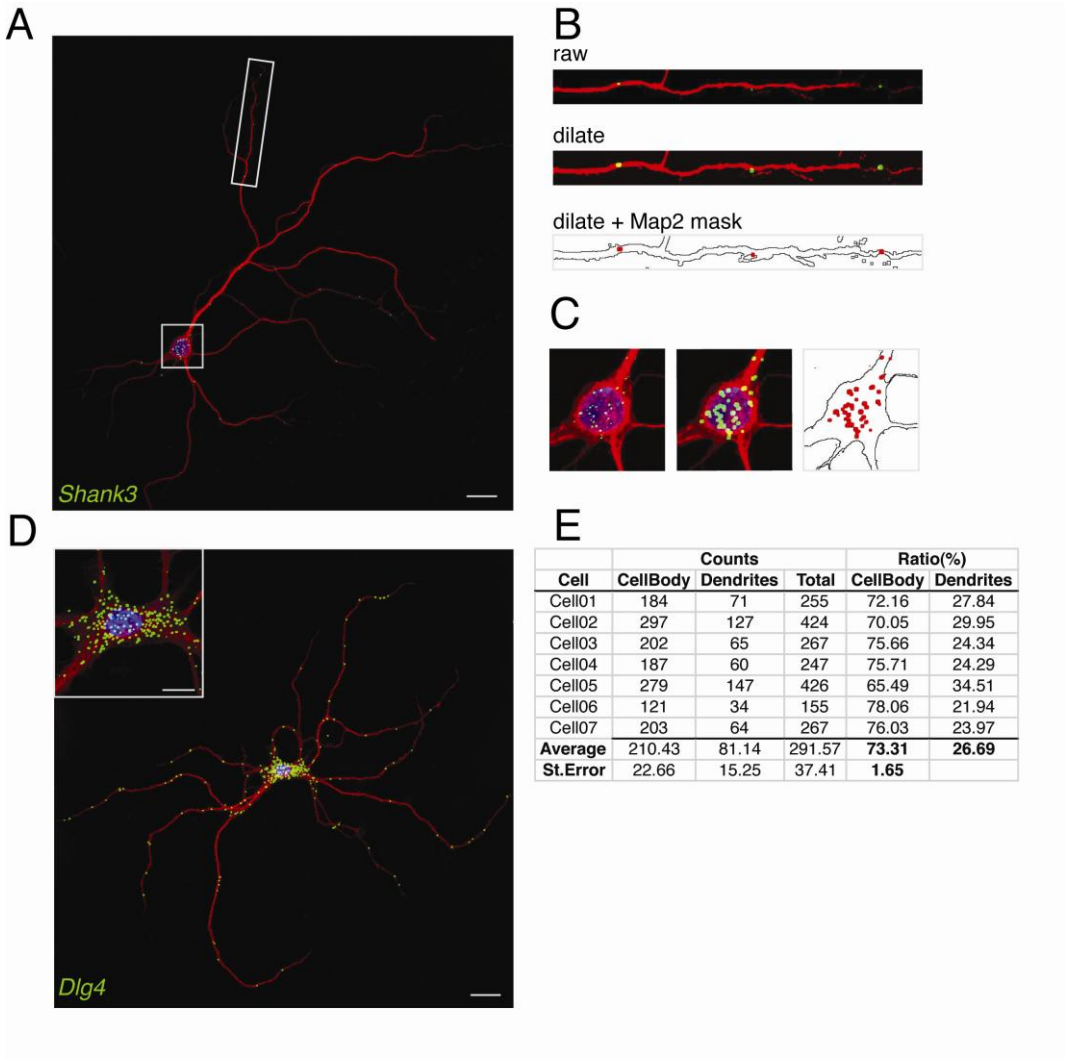


Figure S4, related to Figure 6. Raw images from high-resolution *in situ* hybridization experiments.

All *in situ* hybridization experiments were performed in dissociated hippocampal neurons DIV = 21 (see methods). In the images shown, signal from the in situ hybridization is green. Dendrites were immunostained using an anti-Map2 antibody (red). The nuclei were stained with DAPI (blue). Scale bar = 20 microns. All images were taken using the same imaging conditions. The images depict 12 different transcripts which are present in the dendrites. The red/blue/green/purple circle following the transcript name indicates the membership of the transcript in a cluster with the same color code as shown in Figure 6C.



	Counts			Ratio(%)	
Cell	**CellBody**	**Dendrites**	**Total**	**CellBody**	**Dendrites**
Cell01	184	71	255	72.16	27.84
Cell02	297	127	424	70.05	29.95
Cell03	202	65	267	75.66	24.34
Cell04	187	60	247	75.71	24.29
Cell05	279	147	426	65.49	34.51
Cell06	121	34	155	78.06	21.94
Cell07	203	64	267	76.03	23.97
Average	210.43	81.14	291.57	**73.31**	**26.69**
St.Error	22.66	15.25	37.41	**1.65**	

Figure S5, related to Figure 6. Processing of high-resolution *in situ* hybridization images and relative occupancy of Dlg4 mRNA particles in the soma and dendrite.
(A) Raw image of an *in situ* hybridization experiment for the *Shank3* mRNA (green) in cultured hippocampal neurons (DIV 21). Dendrites were immunostained using an anti-Map2 antibody (red). The nuclei were stained with DAPI (blue). Scale bar = 20 microns.
(B) High magnification (zoom = 3x) versions of a linearized (during analysis) distal dendrite taken from A. Within this set, the raw image is positioned at the top. The middle image (dilate) shows the mRNA puncta dilated twice to increase visibility. In the bottom-most image, the puncta signals were converted to a binary image and were merged with an outline generated from the Map2 signal (dilate + Map2 mask).
(C) High magnification (zoom = 4x) view of the soma in A. Within this set, the raw image is positioned at the left. The middle image (dilate) shows the mRNA puncta dilated twice to increase visibility. In the image at the right, the puncta signals were converted to a binary image and were merged with an outline generated from the Map2 signal (dilate + Map2 mask).
(D) *In situ* hybridization for the *Dlg4* mRNA (green) in cultured hippocampal neurons. Dendrites were immunostained using an anti-Map2 antibody (red). The nuclei were stained with DAPI (blue). Scale bar = 20 microns. The inset shows a magnification for the cell body. Scale bar = 10 microns.
(E) Quantitative estimates of mRNA puncta in the cell body and dendrites from a high resolution *Dlg4 in situ* hybridization experiment. Puncta were counted (see methods) in both compartments. For *Dlg4* ~27% of the total puncta present were detected in the dendrites. Note that the somatic puncta represent the population of mRNA that will be retained in the soma as well as those that will eventually be trafficked to the dendrite.

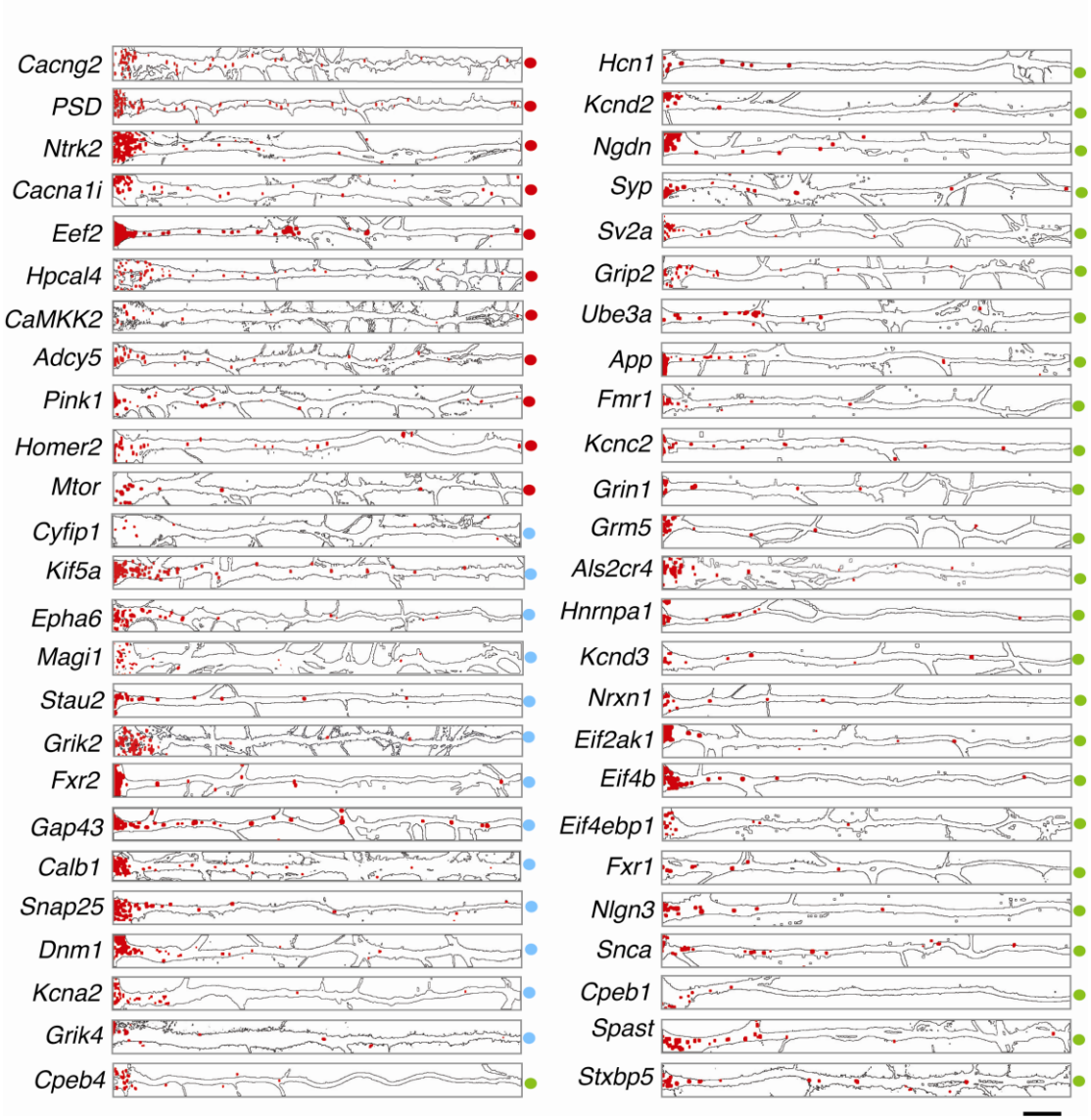
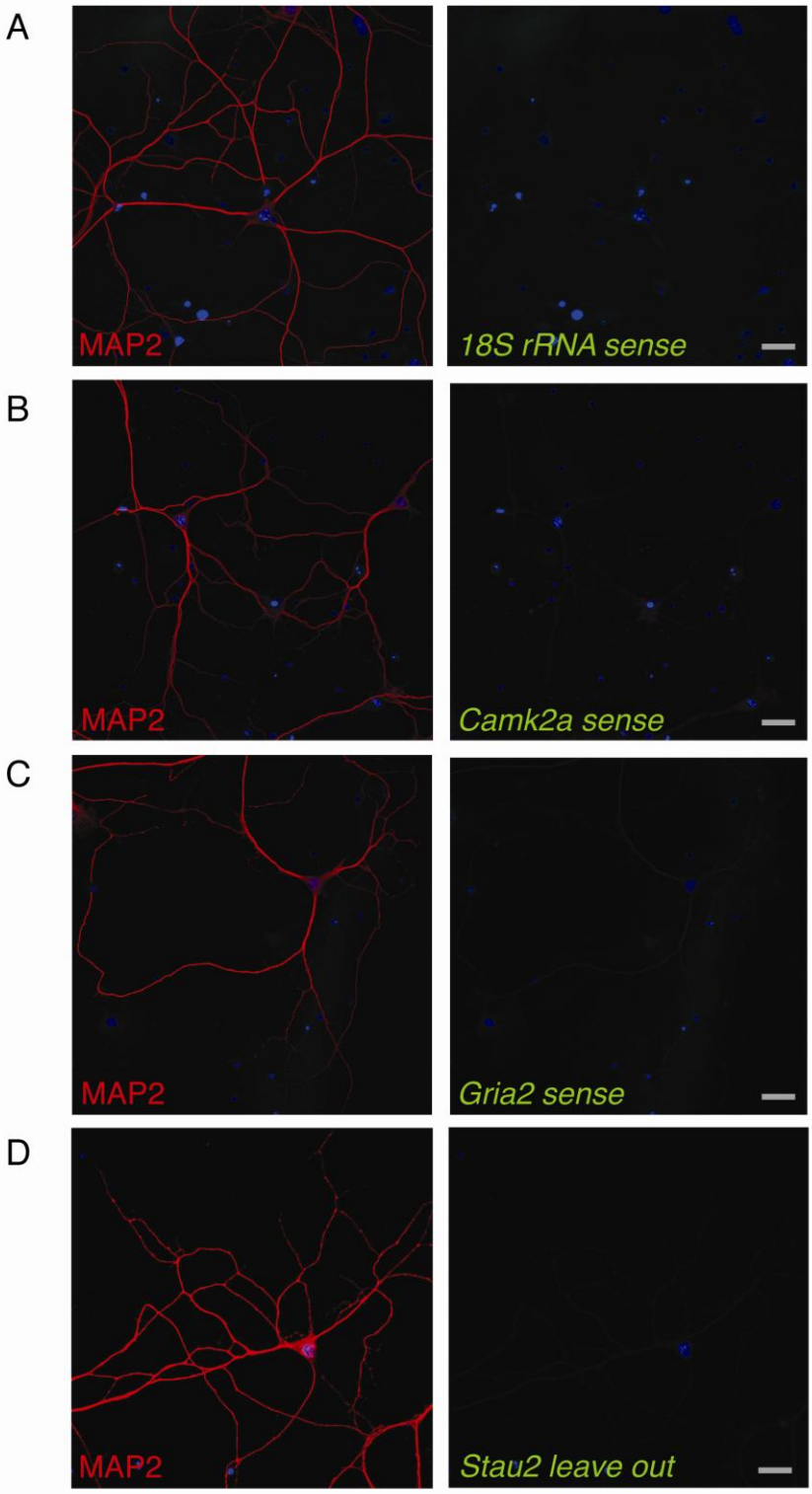


Figure S6, related to Figure 7. High-resolution fluorescent *in situ* hybridization detection of mRNA transcripts in hippocampal dendrites.
FISH signal in individual dendrites from cultured hippocampal neurons linearized for analysis. MAP2 immunostaining was used to generate a mask that outlines the dendritic structure. The red particles show the transcripts for each indicated gene within the dendrites. The particle size has been enlarged to ease visualization. The transcripts vary in their abundance as well as their distribution along the length of the dendrite. The colored dot next to each dendrite represents the cluster identity of each transcript- as indicated in Figure 6. Scale bar = 20 microns.


A
MAP2
18S rRNA sense
B
MAP2
Camk2a sense
C
MAP2
Gria2 sense
D
MAP2
Stau2 leave out

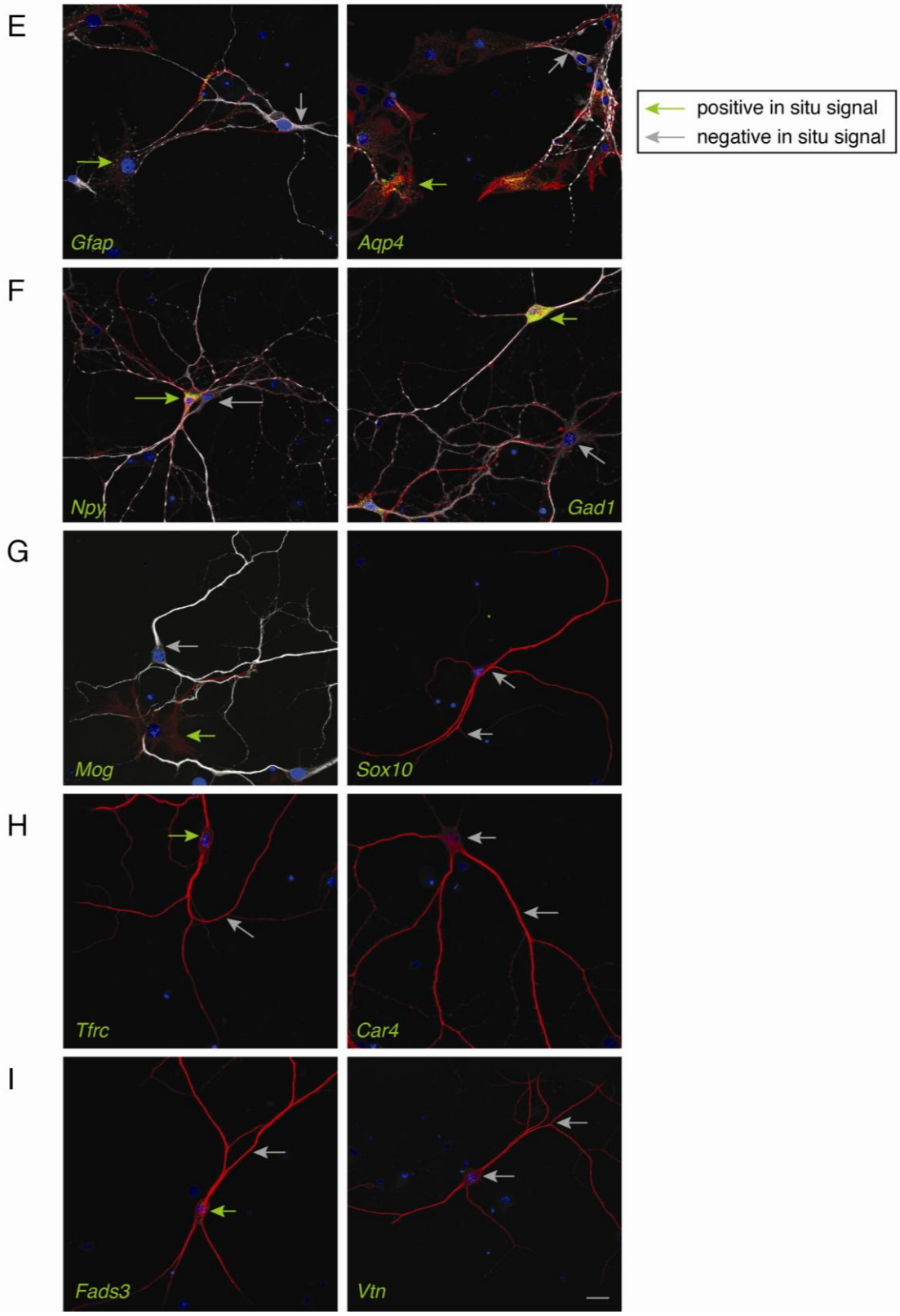

E
Gfap
Aqp4
positive in situ signal
negative in situ signal
F
Npy
Gad1
G
Mog
Sox10
H
Tfrc
Car4
I
Fads3
Vtn

Figure S7, related to Figures 5 and 6. High-resolution in situ hybridization control experiments and verification of filtering process

All *in situ* hybridization experiments were performed in dissociated hippocampal neurons DIV = 21(see methods). In the images shown, signal from the in situ hybridization is green. Dendrites were immunostained using an anti-Map2 antibody (red). The nuclei were stained with DAPI (blue). Scale bar = 20 microns. In all images the first column shows the signal associated with Map2 immunostaining, highlighting the dendritic arbor of neurons. The second column shows the results of *in situ* hybridization.

(A) The sense (2nd column) in situ signal (green) for the 18S-rRNA indicating a very low level of background.

(B) The sense (2nd column) in situ signal (green) for *Camk2a* mRNA, showing very low background levels.

(C) The sense (2nd column) in situ signal (green) for *Gria2* (GluR2, GluA2) mRNA, showing very low background levels.

(D) The no secondary probe control (2nd column) in situ signal (green) for *Stau2* mRNA, showing very low background levels.

(E) Atrocyte-specific genes. Left: *Gfap* mRNA (green) is present in GFAP-positive cells (red; glia) but not in neurons which are MAP2-positive (grey). Right: *Aqp4* mRNA (green) is present in GFAP-positive cells (red) but not in neurons (MAP2-positive, grey).

(F) Interneuron specific genes. Left: *Npy* mRNA (green) is present in GAD-positive cells (interneurons; red) but not in GAD-negative cells (neurons are Map2-positive, grey). Right: *Gad1* mRNA (green) is present in GAD-positive cells (red) but not in GAD- negative cells. Note: all cells present are Map2-positive (neurons, grey).

(G) Oligodendrocytes specific genes. Left: *Mog* mRNA (green) is present in GFAP-positive cells (red) but not in MAP2-positive cells (neurons, grey). Right: *Sox10* mRNA (green) is not present in MAP2-positive (neurons, red) soma or dendrites.

(H) Endothelial cells specific genes. Left: *Tfrc* mRNA (green) is expressed at very low levels in the soma of MAP2-positive cells (neurons, red) but not in the neuropil. Right: *Car4* mRNA (green) is not expressed in MAP2-positive cells (neurons, red).

(I) Pericyte-specific genes. Left: *Fads3* mRNA (green) is expressed at low levels in the soma of MAP2-positive cells (neurons, red) but not in the dendrites. Right: *Vtn* mRNA (green) is not expressed in MAP2-positive cells (neurons, red). Scale bar = 20 microns.